

Discrete Mixing Operators from Boundary Sector Geometry

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Abstract

The CKM and PMNS mixing matrices of the Standard Model lack a first-principles geometric origin. Let F be a finite set of fermion labels; each $f \in F$ is assigned a distinct boundary direction $\xi(f)$ on the sphere at infinity $S^2 = \partial\mathbb{H}^3$ of a compact hyperbolic 3-manifold. Using a Hilbert space of boundary data with a continuous family of normalized vectors $|\xi\rangle$, we define sector subspaces $H_S = \text{span}\{|\xi(f)\rangle : f \in S\}$ for any ordered subset $S \subset F$. The mixing matrix U_{ST} between two sectors S, T of equal size is the matrix of the orthogonal projection $P_T|_{H_S}$ in the Gram-Schmidt orthonormal bases of H_S and H_T . We prove that under the discrete assignment (the set $\Xi = \{\xi(f) : f \in F\}$ is finite and locally rigid) the accessible mixing matrices form a finite set modulo $U(1)^n \times U(1)^n$ rephasings. Consequently, no continuous deformation of the boundary points can produce continuous drift in any rephasing-invariant mixing observable — a structural constraint that any realistic model must satisfy. In companion papers [7, 8], this framework is applied to two arithmetic hyperbolic 3-manifolds, reproducing the CKM and PMNS matrices with Frobenius fitness 0.016482 and 0.005087 respectively.

1 Introduction

Mixing matrices, such as the CKM matrix in the quark sector and the PMNS matrix in the lepton sector, encode the mismatch between flavor and mass eigenstates in the Standard Model [1]. Despite their fundamental importance, their origin remains unexplained by the gauge structure alone. Many attempts invoke discrete flavor symmetries, extra dimensions, or string-inspired constructions [2, 3]. In all such approaches, the mixing matrix ultimately measures the overlap between states associated with different sectors of the theory.

In this paper we take a kinematic, model-independent perspective. We show that if fermions are assigned to distinct boundary directions of a compact hyperbolic 3-manifold, the mixing matrices arising from the overlap of boundary-localized states are determined solely by the discrete boundary data, and form a discrete set modulo sector-local rephasings. A corollary is that no continuous deformation of physical parameters can produce continuous drift in any rephasing-invariant mixing observable.

The construction is deliberately minimal: a Hilbert space of boundary data, a family of boundary-localized states, and a discrete assignment of fermions to boundary directions. No dynamics, no Lagrangian, and no ultraviolet completion are assumed.

In companion papers [7, 8], this framework is applied to two arithmetic hyperbolic 3-manifolds: $M_{\text{PMNS}} = m003(-2, 3)$ (OrientableClosedCensus[1]) and $M_{\text{CKM}} = m006(-5, 2)$ (OrientableClosedCensus[43]). The Borel (N-factor Iwasawa) construction on M_{PMNS} reproduces the PMNS matrix with Frobenius fitness 0.005087 against PDG 2024 values — the global minimum of the construction ($p < 10^{-4}$ vs 50,000 Haar-random unitaries). The K-factor construction on M_{CKM} gives fitness 0.016482 with Cabibbo angle error 0.19%. The present paper

provides the kinematic guarantee that these results are not accidents of continuous parameter tuning.

2 Geometric Setup

Let Γ be a torsion-free discrete group acting properly discontinuously and cocompactly on $\mathbb{H}^3$ by orientation-preserving isometries. The quotient $M = \mathbb{H}^3/\Gamma$ is a compact oriented hyperbolic 3-manifold with boundary at infinity $S^2 = \partial\mathbb{H}^3$.

Definition 1 (Boundary direction). *A boundary direction is a point $\xi \in S^2$. For $\gamma \in \Gamma$, the associated boundary direction is the ideal endpoint of the geodesic ray from a fixed basepoint $o \in \mathbb{H}^3$ through $\gamma \cdot o$.*

Assumption 2 (Discrete sector assignment). *Each fermion flavor $f \in F$ is assigned a unique boundary direction $\xi(f) \in S^2$ by a deterministic, discrete selection rule: the image $\Xi = \{\xi(f) : f \in F\}$ is finite, and the assignment is locally rigid.*

Remark 3 (Geometric realization). *Fix $o \in \mathbb{H}^3$. For each f , let $\gamma_f \in \Gamma$ be the unique element minimizing $d(o, \gamma \cdot o)$ in the assignment class of f (uniqueness follows from proper discontinuity and torsion-freeness [6]). The boundary direction $\xi(f)$ is the ideal endpoint of the geodesic from o through $\gamma_f \cdot o$. In the HFG application [7, 8], γ_f is a short word in $\pi_1(M)$ and $\xi(f) = \hat{n}(\gamma_f)$ is the geodesic axis extracted via the Pauli decomposition of $\log \rho(\gamma_f)$.*

3 Boundary Hilbert Space and Sector Subspaces

Let $\mathcal{H}$ be a separable complex Hilbert space (concretely $L^2(S^2)$).

Definition 4 (Boundary-localized state). *For each $\xi \in S^2$, fix a normalized vector $|\xi\rangle \in \mathcal{H}$ with $\xi \mapsto |\xi\rangle$ continuous, $\langle \xi | \xi \rangle = 1$, and $|\langle \xi | \zeta \rangle| < 1$ for $\xi \neq \zeta$. An explicit family is given by the Poisson kernel $|\xi\rangle \propto P(\cdot, \xi)$ normalized in $L^2(S^2)$ [5].*

Definition 5 (Sector subspace and preferred basis). *A sector S is a finite ordered subset of F . The sector subspace is $H_S = \text{span}\{|\xi(f)\rangle : f \in S\} \subset \mathcal{H}$, with $\dim H_S = |S|$. The preferred ONB $\{|S, 1\rangle, \dots, |S, n\rangle\}$ of H_S is the Gram-Schmidt orthonormalization of $(|\xi(f_1)\rangle, \dots, |\xi(f_n)\rangle)$ with all pivot coefficients real and positive, removing the $U(1)^n$ phase ambiguity.*

4 Mixing Matrices as Overlap Operators

Definition 6 (Mixing matrix). *For sectors S, T of equal cardinality n , the mixing matrix U_{ST} has entries $(U_{ST})_{ji} = \langle T, j | S, i \rangle$.*

Proposition 7 (Unitarity and geometric dependence). *When $H_S = H_T$, $U_{ST} \in U(n)$; in general it is a partial isometry. U_{ST} is determined entirely by $\Xi_{ST} = \{\xi(f) : f \in S \cup T\}$ and the inner product on $\mathcal{H}$: all Gram-Schmidt coefficients are algebraic expressions in $\langle \xi(f) | \xi(g) \rangle$, $f, g \in S \cup T$.*

5 Rephasing Invariance

Physical observables are invariant under $U_{ST} \rightarrow D_T U_{ST} D_S^\dagger$, $D_S, D_T \in U(1)^n$.

Proposition 8 (Rephasing-invariant quantities). *Invariant under sector-local rephasings: (i) moduli $|(U_{ST})_{ji}|^2$; (ii) quartic products $J_{ji,j'i'} = (U_{ST})_{ji}(U_{ST})_{j'i'}\overline{(U_{ST})_{j'i'}}\overline{(U_{ST})_{ji}}$, generalizing the Jarlskog invariant [4]; (iii) $\det U_{ST}$ up to an overall phase. For $n = 3$, $\text{Im } J_{ji,j'i'}$ reduces to J_{CP} .*

6 Stability and Discreteness

Theorem 9 (Stability). *Let $\delta_{\min} = \min_{f \neq f'} d_{S^2}(\xi(f), \xi(f'))$ and $\varepsilon_0 = \delta_{\min}/2$. For any deformation of magnitude $\varepsilon \leq \varepsilon_0$, the perturbed configuration determines the same preferred bases and the same U_{ST} .*

Proof. For $\varepsilon \leq \varepsilon_0$, each $\xi'(f)$ remains closer to $\xi(f)$ than to any $\xi(g)$, $g \neq f$, so labels are preserved and linear independence holds. Gram-Schmidt coefficients depend continuously on inner products; since the discrete assignment rule (Assumption 2) is unchanged, U_{ST} depends only on the unchanged configuration class (by the geometric dependence proposition). $\square$

Theorem 10 (Discreteness of mixing classes). *Under Assumption 2, the set $\mathcal{M}_{ST} = \{U_{ST} : \Xi_{ST} \text{ admissible}\}$ is finite, hence discrete, in $U(n)/(U(1)^n \times U(1)^n)$.*

Proof. The set of admissible Ξ_{ST} is finite. The map $\Xi_{ST} \mapsto U_{ST}$ is continuous, hence sends a finite set to a finite set. Quotienting by the rephasing action preserves finiteness. $\square$

Corollary 11 (No continuous drift). *For deformations of magnitude $\varepsilon \leq \varepsilon_0$, no continuous drift occurs in any rephasing-invariant observable. For larger deformations, the mixing matrix changes only by a discrete jump within $\mathcal{M}_{ST}$.*

Proof. Rephasing-invariant observables are continuous on $U(n)/(U(1)^n \times U(1)^n)$; their restriction to the finite set $\mathcal{M}_{ST}$ takes only finitely many values. $\square$

7 Connection to Hyperbolic Flavor Geometry

The abstract framework is concretely realized in the HFG programme [7, 8, 9] as follows. Set $\Gamma = \pi_1(M)$ with holonomy $\rho : \pi_1(M) \rightarrow \text{PSL}(2, \mathbb{C})$. For a loxodromic element γ , the boundary direction $\xi(\gamma) = \hat{n}(\gamma) \in S^2$ is the geodesic axis extracted via the Pauli decomposition of $\log \rho(\gamma)$, and the inner product becomes the Gaussian kernel $\langle \xi(f) | \xi(g) \rangle = \exp(-\theta_{fg}^2/(2\sigma^2))$, $\theta_{fg} = \arccos |\hat{n}(f) \cdot \hat{n}(g)|$.

For $M_{\text{CKM}} = m006(-5, 2)$ with word triple $\{\text{aaB}, \text{AbA}, \text{AAb}\}$: the homological class collision $[\text{AbA}] = [\text{AAb}] = 2$ in $H_1(M) = \mathbb{Z}/5$ forces $J_{\text{CP}} = 0$ via the No Continuous Drift Corollary, consistent with $|J_{\text{CKM}}| \approx 3 \times 10^{-5}$; the Frobenius fitness is 0.016482 against PDG 2024.

For $M_{\text{PMNS}} = m003(-2, 3)$ with Borel construction on $\{\text{aa}, \text{aaB}, \text{baa}\}$: the distinct classes $[\text{aa}] = 1$, $[\text{aaB}] = 4$, $[\text{baa}] = 3 \pmod{5}$ ensure $J_{\text{CP}} \neq 0$, consistent with large leptonic CP violation; the fitness is 0.005087, the global minimum of the construction.

8 Conclusions

Under the discrete boundary assignment, the set of admissible mixing matrices is finite modulo rephasings (Theorem 10), and bounded deformations cannot produce continuous drift in physical observables (Theorem 9). This provides a model-independent kinematic explanation for why the CKM and PMNS mixing matrices cannot vary continuously: they are selected from a discrete geometric set determined by the topology of the underlying manifold.

Data Availability

No research data generated. Companion scripts: <https://github.com/drmlgentry/hyperbolic-flavor-scan>

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